

NAM HOAI LE

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RESEARCH INTERESTS

Mathematical Cryptography, Number Theory, and Lattices

EDUCATION

Florida Atlantic University

Ph.D. in Mathematics, Current cumulative GPA: 4.00/4.00

Boca Raton, Florida, USA

Aug 2024 – Present

Ho Chi Minh City University of Science

M.S. in Algebra and Number Theory, GPA: 8.51/10.0

Ho Chi Minh City, Vietnam

Dec 2022

Thesis: Well-rounded twists of ideal lattices from imaginary quadratic fields.

Thesis grade: 10/10.

Supervisors: Assoc. Prof. Ha Tran and Dr. Le Van Luyen.

Ho Chi Minh City University of Education

B.S. in Mathematics, GPA: 3.45/4.0

Ho Chi Minh City, Vietnam

Jun 2019

Thesis: Spectrum of a ring.

Thesis grade: 10/10.

Supervisor: Assoc. Prof. My Vinh Quang.

PUBLICATIONS AND PREPRINTS

- [1] **Optimal Bucket Set Construction for Multi-scalar Multiplication with Endomorphisms.**
NAM HOAI LE and Francesco Sica.
To appear in ASIACRYPT 2026.
- [2] **The exceptional set in Cassels's theorem on small cyclotomic integers.**
Jitendra Bajpai, Srijan Das, Kiran S. Kedlaya, NAM HOAI LE, Meghan Lee, Antoine Leudière, and Jorge Mello.
Journal für die reine und angewandte Mathematik (Crelle's Journal), accepted for publication.
- [3] **Well-rounded twists of the ring of integers in cyclic cubic fields.**
NAM HOAI LE, Dat T. Tran, David Karpuk, and Ha T. N. Tran.
Preprint; under major revision.
- [4] **Well-rounded ideal lattices of cyclic cubic and quartic fields.**
Dat T. Tran, NAM HOAI LE, and Ha T. N. Tran.
Communications in Mathematics, 31(2), 209–250, 2023.

[5] **Well-rounded twists of ideal lattices from imaginary quadratic fields.**

NAM HOAI LE, Dat T. Tran, and Ha T. N. Tran.

Journal of Algebra and Its Applications, 21(7), 2250133, 2022.

TALKS AND PRESENTATIONS

- **Optimal Bucket Set Construction for Multi-scalar Multiplication with Endomorphisms.**
ASIACRYPT 2026, Hong Kong.
December 2026. Scheduled presentation.
- **Optimal Bucket Set Construction for Multi-scalar Multiplication with Endomorphisms.**
Crypto Café, Department of Mathematics and Statistics, Florida Atlantic University.
September 2026.
- **The Algorithms of Shor.**
Two-part presentation, Quantum Algorithms Seminar, Department of Mathematics and Statistics, Florida Atlantic University.
March 2026.

EMPLOYMENT

Florida Atlantic University

Graduate Teaching Assistant

Boca Raton, Florida, USA

Aug 2024 – Present

Ho Chi Minh City University of Science

Adjunct Lecturer

Ho Chi Minh City, Vietnam

Feb 2024 – Jun 2024

VietQuant

Quantitative Researcher

Ho Chi Minh City, Vietnam

Apr 2020 – Oct 2023

TEACHING EXPERIENCE

Florida Atlantic University

- MAS 4304/5311 – Introductory Abstract Algebra 1
Lecture & Recitation Instructor (Graduate Teaching Assistant), Fall 2026.
- MAC 2233 – Methods of Calculus
Instructor of Record, Spring 2026.
- MAC 2311 – Calculus with Analytic Geometry 1
Grader, Spring 2026.
- MGF 1130 – Math Thinking in Context 1
Instructor of Record, Fall 2025.
- MAC 2312 – Calculus with Analytic Geometry 2
Grader, Spring 2025 and Fall 2024.
- Math Learning Center
Tutor, Fall 2024 – Present.

Ho Chi Minh City University of Science

- MTH10324 – Algorithmic Number Theory
Adjunct Lecturer, Spring 2024.

CONFERENCES, WORKSHOPS, AND RESEARCH SCHOOLS

- **PKC 2026**
The International Conference on Practice and Theory in Public Key Cryptography.
West Palm Beach, Florida, USA, May 26–28, 2026.
- **Applied Algebra Days Workshop**
Florida Atlantic University.
October 11–12, 2025.
- **Preliminary Arizona Winter School 2025**
Mathematical Cryptography and Algorithms in Number Theory.
Online, September 22 – November 21, 2025.
- **Graduate Workshop on Linear Algebra over Finite Fields & Applications**
ICERM, Brown University.
August 18–29, 2025.
- **Rethinking Number Theory 6 (RNT 6)**
Online.
June 16–27, 2025.
- **CIMPA School on Applied Number Theory**
Ho Chi Minh City University of Education.
June 3–14, 2024.
- **SEAMS School on Number Theory and Applications**
Industrial University of Ho Chi Minh City, Vietnam.
June 12–22, 2023.
- **CIMPA Research School on Group Actions in Arithmetic and Geometry**
Gadjah Mada University, Indonesia.
February 17–28, 2020.

TECHNICAL SKILLS

Programming: Python

Mathematical Software: SageMath, PARI/GP

Typesetting: $\text{\LaTeX}$

LANGUAGES

Vietnamese (Native); English (Professional proficiency)